

# Blood & platelets — the science behind PRP

బ్లడ్ & ప్లేట్‌లెట్స్ — PRP సైన్స్



## TODAY'S OBJECTIVES · ఈ రోజు లక్ష్యాలు

- ✦ Explain what PRP is and how it stimulates hair growth
- ✦ Understand blood components and what the centrifuge separates
- ✦ Prepare the theory you need before touching a tube

## THEORY · థియరీ

### Blood components

Whole blood = plasma (55%), buffy coat (platelets + white cells) and red cells (45%). Spinning separates them by density. Platelet-rich plasma is plasma with platelets concentrated 3–6× above baseline. Platelets contain alpha granules full of growth factors — PDGF, VEGF, EGF, IGF-1, TGF- $\beta$ .

### How it works on hair

Injected around the follicles, the growth factors increase dermal papilla cell activity and blood supply, prolong anagen, and reduce miniaturisation. Effect: less shedding by 4–6 weeks, visible new growth by 3 months, density improvement by 5–6 months. It supports — it does not replace — the doctor's medical therapy, and it cannot regrow follicles that are already scarred.

### Tubes and activation

PRP tubes contain an anticoagulant (ACD-A or sodium citrate) and often a separating gel. Handle gently — rough shaking damages platelets. Some protocols activate with calcium chloride; many hair protocols use non-activated PRP. Always follow the doctor's chosen protocol exactly, including spin speed and time.



### Hands-on today

- ✦ Identify tube types, anticoagulants and their uses
- ✦ Watch a full spin and identify the layers in the tube
- ✦ Write the clinic's exact PRP protocol from observation



### Safety & contraindications

- ✦ Never use a tube past its expiry or with a broken seal
- ✦ Label every tube with the client's name immediately after draw — no exceptions

### KEY POINTS TO REMEMBER · గుర్తుంచుకోవలసినవి

- ✦ Platelets concentrated 3–6x
- ✦ Growth factors: PDGF, VEGF, IGF-1
- ✦ Shedding reduces by 4–6 weeks
- ✦ Follow the clinic's spin protocol exactly



### Homework

Draw a labelled tube after centrifugation and write which layer is used.



### Grooming & professional tip

One mislabelled tube can mean someone else's blood in a client. Label at the chair, in front of the client, every single time.

### SELF-CHECK · సెల్ఫ్ చెక్

- ✦ Did I follow hygiene and consent for every client today?
- ✦ Which step did I need help with — and what will I drill tonight?
- ✦ Did I write my records and settings before leaving?

Trainer sign: \_\_\_\_\_ Trainee sign: \_\_\_\_\_